A Real Time Research Project Report

On

Smart Emergency Triage

Submitted in partial fulfilment for the Degree of B. Tech.

In

COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

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(An Autonomous Institution)
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Aziz Nagar Gate, C.B. Post, Hyderabad-500075

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CERTIFICATE

This is to certify that the project report entitled **"Smart Emergency Triage"** submitted by **Charan Teja [24911A66B6] ,G Naresh [25915A6607], M Venkatesh [25915A6612]** to Vidya Jyothi Institute of Technology(An Autonomous Institution), Hyderabad, in partial fulfilment for the award of the degree of **B. Tech. in** Computer Science and Engineering (Artificial Intelligence & Machine Learning),a bonafide record of project work carried out by us under my supervision. The contents of this report, in full or in parts, have not been submitted to any other Institution or University for the award of any degree.

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DECLARATION

We declare that this project report titled **Smart Emergency Triage** submitted in partial fulfilment of the degree of Bachelor of Technology in **Computer Science and Engineering(Artificial Intelligence & Machine Learning)** is a record of original work carried out by us under the supervision of **Dr.B.Nanditha**, and has not formed the basis for the award of any other degree or diploma, in this or any other Institution or University. In keeping with the ethical practice of reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

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Finally, we would like to express our deepest gratitude to our **parents** whose unwavering love, guidance, and support have been the foundation of our success and aspirations.

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Abstract

The Smart Emergency Triage System is a web-based decision-support system designed to improve the speed and quality of emergency medical care. In emergency situations, every second is important, and quick decisions are needed to save lives. This system helps ambulance staff and hospital teams by automatically analyzing patient conditions and supporting fast decision-making.

The system collects important patient health data such as heart rate, blood pressure, respiratory rate, oxygen saturation (SpO2), body temperature, blood glucose level, and consciousness level (GCS). Using this information, the system calculates a severity score for each patient. Based on this score, patients are classified into three priority levels: **High, Medium, or Low**. This helps medical staff identify critical patients who need immediate treatment.

In addition to triage classification, the system also includes a smart hospital recommendation feature. It checks hospital details such as available beds, ICU facilities, medical specialization (like cardiology or trauma care), and distance from the patient location. Based on these factors, the system suggests the most suitable hospital for each patient. This reduces confusion and saves valuable time during emergencies.
The system also provides a simple dashboard where hospital staff can view incoming patient details in real time. They can easily monitor, filter, and manage patient queues, which improves coordination between ambulance teams and hospitals.

The application is developed using basic web technologies such as HTML, CSS, and JavaScript, making it lightweight and easy to use without complex installation or advanced backend systems. It runs in a browser and is designed to be responsive for both desktop and mobile devices.

Overall, the Smart Emergency Triage System helps improve emergency healthcare by combining patient data analysis, priority classification, and hospital recommendation into one platform. It reduces delay in treatment, improves communication between emergency responders and hospitals, and supports better patient outcomes in critical situations. In the future, it can be improved with features like real-time hospital API integration, machine learning-based predictions, cloud storage, and voice-based input for faster data entry.

CHAPTER No.	TITLE	PAGE NO.
	Certificate	2
	Declaration	3
	Acknowledgement	4
	Abstract	5
	List of Figures (Tabular Form)	6
	List of Tables (Tabular Form)	7
1	**INTRODUCTION**	8
	1.1 Introduction	8
	1.2 Need for the Project	8
	1.3 Objectives of the Project	8
	1.4 Scope of the Project	9
	1.5 Problem Statement	9
	1.6 Advantages of the Proposed System	9
2	**LITERATURE SURVEY**	10
	2.1 Introduction	10
	2.2 Traditional Ambulance Systems	11
	2.3 Existing Triage Systems	11
	2.4 GPS-Based Ambulance Tracking Systems	11
	2.5 Artificial Intelligence in Healthcare	11
	2.6 Hospital Recommendation Systems	12
	2.7 Comparative Analysis	12
3	**SYSTEM ANALYSIS**	13
	3.1 Existing System	13
	3.2 Proposed System	13
	3.3 Feasibility Study	13
4	**SYSTEM REQUIREMENTS**	14
	4.1 Software Requirements	14
	4.2 APIs Used	14
	4.3 Hardware requirements	14
5	**SYSTEM DESIGN**	15
	5.1 System Architecture	15
	5.2 Work Flow	16
	5.3 Modules	17
6	**IMPLEMENTATION**	19
	6.1 Technologies Used	19
	6.2 AI-Based Triage Implementation	19
	6.3 Distance Calculation	20
	6.4 Hospital Ranking Algorithm	20
	6.5 Route Navigation	21
7	**RESULTS**	22
	7.1 Result	22
8	**TESTING**	26
	8.1 Introduction	26
	8.2 Types of Testing	26
	8.3 Test Cases	27

9	**CONCLUSION & FUTURE ENHANCEMENTS**	28
	9.1 Conclusion	28
	9.2 Future Enhancements	28
	References	29

List of Figures (Tabular Form)

Fig. No.	Title	Page No.
1.	TriageFLOW Architecture Diagram	15
2.	TriageFLOW Home Page (Screenshot)	22
3.	Patient Vital Signs Input Screen (Screenshot)	23
4.	Nearby Hospitals with Map (Screenshot)	24
5.	Hospital Triage Dashboard (Screenshot)	25

List of Tables (Tabular Form)

Table No.	Title	Page No.
1.	Comparative Analysis (Existing System vs Limitations)	13
2.	Software Requirements	14
3.	AI Triage Example Conditions	19

Chapter 1
Introduction

1.1 Introduction

Healthcare emergency management is one of the most critical areas in modern society. During emergency situations such as cardiac arrest, stroke, trauma, accidents, respiratory failure, and severe infections, every second plays an important role in saving a patient's life. Traditional ambulance systems often depend on manual decision-making, which may lead to delays in identifying patient severity, selecting hospitals, and reaching medical facilities.

In many emergency situations, ambulance staff must quickly determine the seriousness of the patient's condition and decide which hospital is best suited for treatment. However, traditional systems do not provide intelligent assistance for emergency classification, hospital selection, or navigation support. This may result in delays, traffic-related problems, incorrect hospital allocation, and inefficient emergency handling.To overcome these challenges, the proposed project "TriageFLOW: AI-Powered Smart Emergency Triage and Ambulance Routing System" introduces an intelligent emergency healthcare platform that combines Artificial Intelligence, GPS tracking, hospital recommendation systems, and route optimization technologies.

The system performs AI-based triage analysis using patient vital signs and symptoms. Based on the entered data, the system predicts patient severity and classifies the emergency level into LOW, MEDIUM, HIGH, or CRITICAL categories. The system then identifies nearby hospitals using real-time GPS location and recommends the best hospital according to the patient's condition and h

ospital capabilities.The project also integrates live map navigation and ETA calculation using routing services. Ambulance staff can preview routes, navigate to hospitals, monitor incoming patients, and manage emergency workflows through a centralized dashboard.

The main aim of TriageFLOW is to improve emergency response efficiency, reduce delays during patient transportation, and assist healthcare professionals in making faster and smarter decisions.

1.2 Need for the Project

Emergency medical services are highly time-sensitive, and delays in treatment can lead to serious complications or even loss of life. Many existing ambulance systems still rely on manual procedures for patient severity identification, hospital allocation, route planning, and communication between ambulances and hospitals. These systems lack intelligent automation and real-time analysis capabilities, which reduces the efficiency of emergency response services. During critical situations, ambulance staff may face challenges such as selecting the wrong hospital, traffic-related delays, lack of information about nearby emergency facilities, absence of live patient monitoring, and delayed decision-making, all of which can negatively affect patient care and survival chances.

1.3 Objectives of the Project

The main objective of TriageFLOW is to develop an AI-based emergency triage system that can analyze patient conditions and classify patients according to severity levels.

The system also aims to provide real-time GPS-based ambulance tracking, dynamically identify nearby hospitals, and recommend the most suitable hospital based on the patient's medical condition, emergency facilities, and distance.

1.4 Scope of the Project

The scope of TriageFLOW includes intelligent emergency healthcare management and smart ambulance routing systems. The project can be effectively used in ambulance services, emergency healthcare centers, smart city healthcare systems, trauma care management, and emergency response networks to improve emergency handling, hospital coordination, and patient care during critical situations.

1.5 Problem Statement

Traditional emergency systems face several problems such as manual triage analysis, delayed hospital selection, lack of intelligent routing, no real-time ambulance tracking, and poor coordination between hospitals and ambulances. These issues reduce emergency response efficiency and increase patient risk during critical situations. TriageFLOW aims to solve these problems using AI-based emergency assessment, smart hospital ranking, GPS-based hospital search, and live ambulance navigation**.**

1.6 Advantages of the Proposed System

The proposed system provides several advantages:

Faster Emergency Response
The system quickly analyzes patient conditions and helps ambulance staff make faster decisions. This reduces delays during emergency situations and improves patient survival chances.

AI-Assisted Triage Analysis
TriageFLOW uses AI-based analysis to predict patient severity using vital signs such as heart rate, blood pressure, SpO2, and symptoms. It helps reduce human errors and improves emergency assessment accuracy.

Smart Hospital Recommendation
The system automatically recommends the most suitable hospital based on patient condition, ICU availability, emergency facilities, distance, and estimated arrival time.

Reduced Ambulance Delays
Using intelligent route planning and live navigation, the system helps ambulances avoid traffic and select faster routes, reducing transportation delays.

Improved Patient Management
Doctors and ambulance staff can monitor patient condition more effectively through continuous vital sign analysis and emergency status updates.

Real-Time GPS Tracking
The system tracks ambulance location in real time using GPS technology, helping hospitals and emergency teams coordinate better during patient transport.

Chapter 2
Literature Survey

2.1 Introduction

A literature survey is an important phase in software development projects. It helps in understanding existing systems, technologies, research works, and limitations. This chapter discusses various emergency healthcare systems, ambulance tracking systems, AI-based healthcare applications, and routing technologies related to the proposed project.

International Journal of Medical Informatics

- AI-driven triage systems analyze **vital signs, symptoms, and patient history** to quickly identify and prioritize critical patients.
- Traditional emergency triage methods are often **manual, time-consuming, and subjective**, which can lead to delays and inconsistent decision-making.
- Machine learning and AI techniques improve **accuracy and speed of patient classification**, reducing human error in emergency care.
- AI-based emergency systems help in **reducing waiting time, improving patient flow, and optimizing hospital resource management** such as beds, staff, and ICU allocation.

BMC Health Services Research [1]

- Efficient **patient flow management in emergency departments** is essential to reduce overcrowding, long waiting times, and delays in treatment.
- Emergency departments face key challenges such as **limited resources, poor coordination, and inefficiencies in patient handling**, which negatively affect service quality.
- The review highlights that improving **workflow organization and hospital coordination systems** can significantly enhance emergency care performance.

BMJ Open[4]

- Emergency triage systems show **varying levels of performance in accurately prioritizing patients**, depending on the method used across different healthcare settings.
- The study highlights that existing triage tools can lead to **inconsistencies in patient classification**, affecting the speed and quality of emergency care.
- Overall, the meta-analysis suggests a need for **improved and standardized triage systems** to enhance accuracy and reliability in emergency departments.

Scandinavian Journal of Trauma, Resuscitation and Emergency Medicine[5]

- Emergency department triage systems use **different structured scales** to categorize patients based on the severity of their condition.
- These triage scales consist of multiple **clinical components such as vital signs, symptoms, and urgency levels**, but they are not always consistent across systems.
- The study highlights that variations in triage methods can lead to **differences in patient prioritization and affect the efficiency of emergency care delivery**

JAMA Network Open[7]

- Artificial intelligence in emergency triage shows **high potential to improve patient prioritization and decision-making**, especially in fast-paced emergency environments.
- Despite its promise, AI-based triage systems still face **limitations such as lack of full clinical validation, data quality issues, and implementation challenges in real hospital settings**.
- The study emphasizes that more research and development are needed to ensure **safe, reliable, and clinically acceptable AI-driven triage systems**.

2.2 Traditional Ambulance Systems

Traditional ambulance systems mainly depend on manual communication between ambulance staff and hospitals. In these systems, ambulance drivers usually select hospitals based on personal experience or general availability without intelligent technological assistance. Such systems have several limitations, including the absence of automated patient severity analysis, real-time hospital ranking, and live route optimization. Due to delayed emergency handling and lack of intelligent decision-making support, these systems can increase patient risk and reduce the efficiency of emergency healthcare services during critical situations.

2.3 Existing Triage Systems

Triage systems are used in hospitals to classify patients based on the severity of their medical condition. Existing triage systems mainly depend on manual assessment and observations by medical staff. Patients are generally categorized into low, medium, high, and critical priority levels. However, these systems have several limitations such as manual processing, chances of human error, time-consuming analysis, and lack of AI-based prediction support. The proposed TriageFLOW system overcomes these issues by automating triage classification using AI-based rules and intelligent emergency assessment.

2.4 GPS-Based Ambulance Tracking Systems

Modern healthcare systems increasingly use GPS technology for ambulance tracking and emergency navigation. GPS helps monitor ambulance movement, identify locations, estimate arrival time, and provide traffic-aware route navigation. Existing GPS systems mainly support vehicle tracking, route guidance, ETA estimation, and traffic monitoring. TriageFLOW enhances these capabilities by combining GPS tracking with intelligent healthcare decision-making, allowing smarter hospital selection and faster emergency response.

2.5 Artificial Intelligence in Healthcare

Artificial Intelligence is widely used in healthcare applications such as disease prediction, medical diagnosis, patient monitoring, emergency analysis, and healthcare automation. AI-based systems help improve accuracy, reduce manual effort, and support faster decision-making in medical environments. The proposed TriageFLOW system uses rule-based AI techniques for emergency triage analysis and patient condition prediction to improve emergency response and healthcare management.

2.6 Hospital Recommendation Systems

Hospital recommendation systems help patients identify suitable hospitals based on healthcare requirements. Existing systems mainly consider factors such as distance, hospital ratings, and facility availability. However, these systems have limitations because they do not support emergency-based prioritization, patient severity analysis, or real-time ambulance routing. TriageFLOW improves hospital recommendation by considering important factors such as patient severity level, ICU support, emergency facilities, distance, and live ambulance navigation to provide faster and smarter hospital selection during emergencies.

2.7 Comparative Analysis

Existing System	**Limitations**
Manual Triage System	Slow emergency classification
Traditional Ambulance System	No AI assistance
GPS Tracking Systems	No hospital recommendation
Hospital Recommendation Systems	No emergency analysis
Static Navigation Systems	No live patient monitoring

Chapter 3
System Analysis

3.1 Existing System

Existing emergency healthcare systems are mostly manual and disconnected, making emergency response slower and less efficient. Ambulance staff often depend on personal judgment for patient management and hospital selection, which can lead to delays and incorrect decisions during critical situations. These systems lack intelligent severity prediction, live hospital ranking, and automated routing support. In addition, poor coordination between hospitals and ambulances, traffic-related delays, and the absence of proper patient monitoring systems further reduce the quality of emergency healthcare services and increase dependency on human intervention.

3.2 Proposed System

The proposed system introduces an intelligent emergency healthcare platform that automates patient assessment and hospital recommendation during emergency situations. TriageFLOW uses AI-based triage analysis to evaluate patient conditions and classify severity levels accurately. The system also provides real-time GPS tracking for ambulance monitoring, nearby hospital search using location services, and smart hospital ranking based on factors such as patient condition, ICU availability, emergency support, distance, and estimated arrival time.

3.3 Feasibility Study

3.3.1 Technical Feasibility

The project is technically feasible because it uses modern web technologies such as:

- HTML
- CSS
- JavaScript
- GPS APIs
- Routing APIs
- Browser local storage

3.3.2 Economic Feasibility

The project is cost-effective because it uses open-source APIs and does not require expensive hardware components. This reduces development and implementation costs while still providing efficient emergency healthcare and ambulance navigation services.

CHAPTER 4

System Requirements

4.1 Software Requirements

Software	Purpose
HTML5	Frontend structure
CSS3	Styling
JavaScript	Application logic
VS Code	Development tool
Google Chrome	Browser testing

4.2 APIs Used

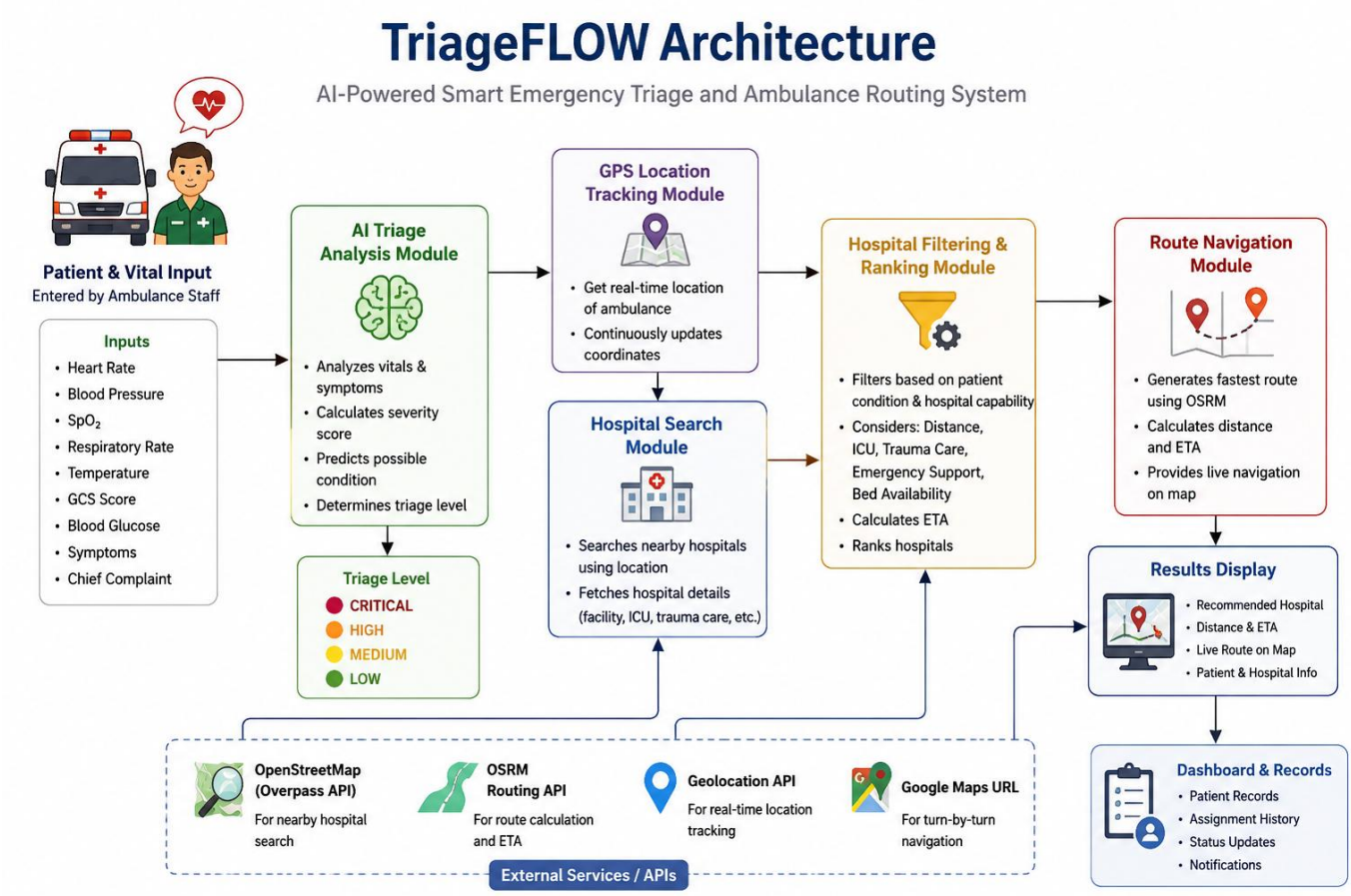
- OpenStreetMap API
- Overpass API
- OSRM Routing API
- Geolocation API
- Google Maps URL API

4.3 Hardware Requirements

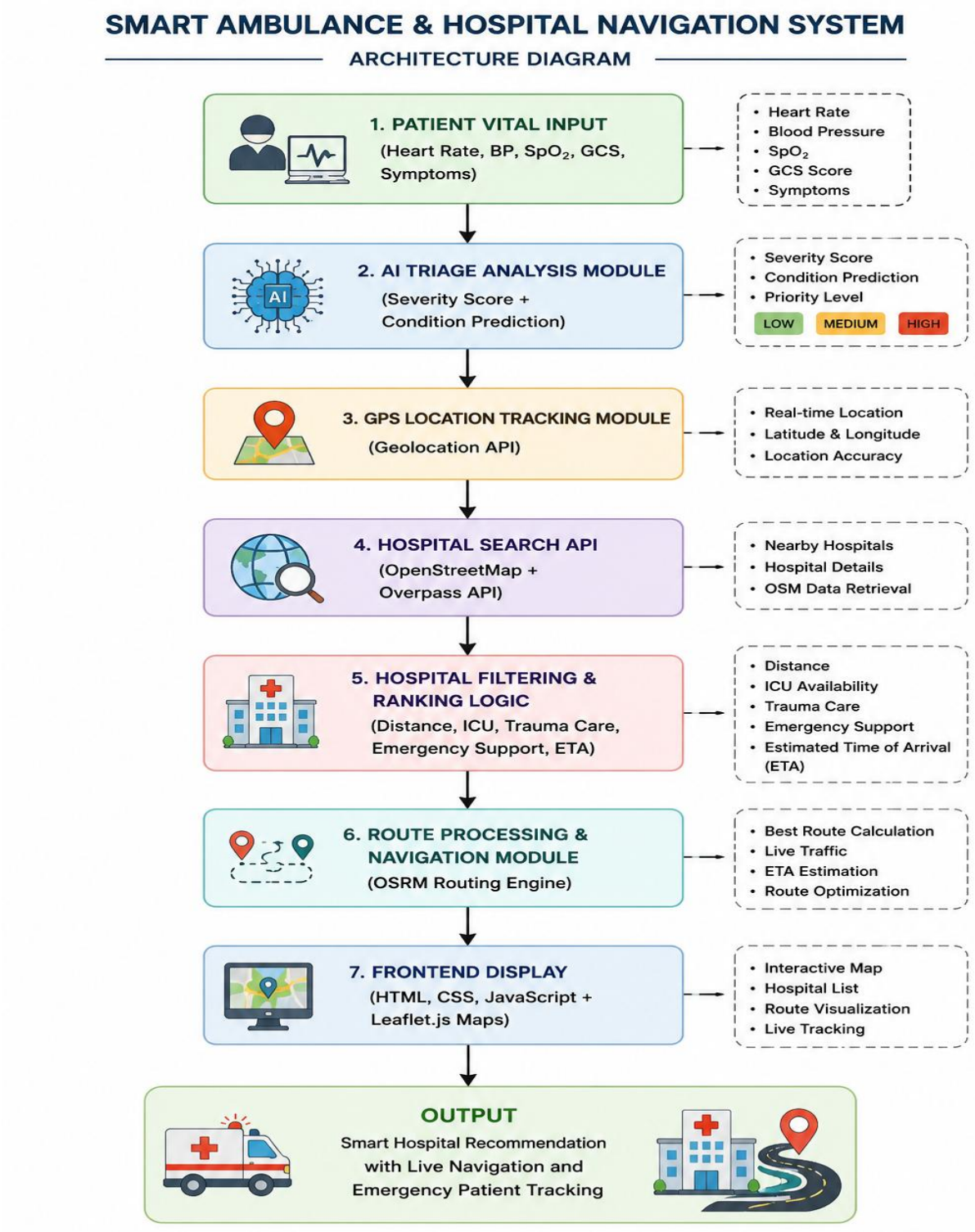
- **Processor:** Intel Core i3 or above
- **RAM:** 4 GB minimum
- **Storage:** 100 GB Hard Disk / SSD
- **System Type:** 64-bit Computer
- **Internet Connection:** Required for Maps and APIs
- **GPS Support:** Mobile GPS or Location Services
- **Input Devices:** Keyboard and Mouse

CHAPTER 5
System Design

5.1 System Architecture



5.2 Work Flow


SMART AMBULANCE & HOSPITAL NAVIGATION SYSTEM
ARCHITECTURE DIAGRAM
1. PATIENT VITAL INPUT
(Heart Rate, BP, SpO2, GCS, Symptoms)
• Heart Rate
• Blood Pressure
• SpO2
• GCS Score
• Symptoms
2. AI TRIAGE ANALYSIS MODULE
(Severity Score + Condition Prediction)
AI
• Severity Score
• Condition Prediction
• Priority Level
LOW
MEDIUM
HIGH
3. GPS LOCATION TRACKING MODULE
(Geolocation API)
• Real-time Location
• Latitude & Longitude
• Location Accuracy
4. HOSPITAL SEARCH API
(OpenStreetMap + Overpass API)
• Nearby Hospitals
• Hospital Details
• OSM Data Retrieval
5. HOSPITAL FILTERING & RANKING LOGIC
(Distance, ICU, Trauma Care, Emergency Support, ETA)
• Distance
• ICU Availability
• Trauma Care
• Emergency Support
• Estimated Time of Arrival (ETA)
6. ROUTE PROCESSING & NAVIGATION MODULE
(OSRM Routing Engine)
• Best Route Calculation
• Live Traffic
• ETA Estimation
• Route Optimization
7. FRONTEND DISPLAY
(HTML, CSS, JavaScript + Leaflet.js Maps)
• Interactive Map
• Hospital List
• Route Visualization
• Live Tracking
OUTPUT
Smart Hospital Recommendation with Live Navigation and Emergency Patient Tracking

5.2 Modules

Patient Vital Input Module

This module collects patient medical information and emergency details entered by ambulance staff. It includes vital parameters such as heart rate, blood pressure, respiratory rate, oxygen saturation (SpO2), temperature, blood glucose level, Glasgow Coma Scale (GCS), symptoms, and chief complaints.

Functions:

- Accepts patient emergency details
- Validates medical input values
- Stores patient information temporarily
- Sends data for triage analysis

AI Triage Analysis Module

This module analyzes patient vital signs and predicts the severity of the medical condition using rule-based AI logic. Based on emergency symptoms and abnormal vital values, the system categorizes patients into LOW, MEDIUM, HIGH, or CRITICAL priority levels.

Functions:

- Calculates triage score
- Predicts emergency conditions
- Detects abnormal vitals
- Generates emergency alerts
- Suggests urgency level

GPS Location Tracking Module

This module tracks the real-time location of the ambulance using browser geolocation services. It continuously updates ambulance coordinates for accurate hospital search and navigation.

Functions:

- Retrieves live GPS coordinates
- Tracks ambulance movement
- Updates current location in real time
- Improves nearby hospital detection

Hospital Search API Module

This module communicates with OpenStreetMap and Overpass APIs to identify nearby hospitals and emergency healthcare centers based on ambulance location.

Functions:

- Searches nearby hospitals
- Retrieves hospital location data
- Fetches hospital details
- Supports live emergency hospital discovery

Hospital Filtering & Ranking Module

This module filters and ranks hospitals according to patient condition, hospital facilities, distance, emergency capabilities, ICU availability, trauma support, and estimated travel time.

Functions:

- Filters hospitals based on emergency needs
- Calculates distance and ETA
- Matches hospital capabilities
- Ranks hospitals intelligently

Route Navigation Module

This module provides live navigation and route mapping using OSRM routing services and Leaflet.js maps.

Functions:

- Generates fastest driving routes
- Displays live map navigation
- Calculates route distance and travel time
- Supports Google Maps navigation integration

Frontend Display Module

This module handles user interaction and displays all emergency information through a responsive web interface.

Functions:

- Displays patient details
- Shows triage results
- Displays nearby hospitals on maps
- Provides navigation controls
- Shows live emergency updates

Patient Record Management Module

This module stores emergency patient assignment records and maintains ambulance-to-hospital transfer history using browser local storage.

Functions:

- Saves patient records
- Tracks hospital assignments
- Maintains emergency logs
- Displays dashboard history and notifications

CHAPTER 6

Implementation

6.1 Technologies Used

HTML

HTML5 is used to create the basic structure and layout of the TriageFLOW application. It is used for designing web pages such as the home page, ambulance dashboard, patient input forms, hospital recommendation page, and navigation interface. HTML elements are also used to display patient details, emergency alerts, hospital information, and map containers.

CSS

CSS3 is used for designing and styling the user interface of the system. It improves the visual appearance of the application by adding colors, layouts, buttons, cards, tables, animations, and responsive design features. CSS helps create a user-friendly emergency dashboard and improves readability during emergency situations.

JavaScript

JavaScript is used to add dynamic functionality and interactivity to the application. It handles patient data processing, emergency severity calculations, API integration, hospital filtering, GPS location fetching, and real-time updates. JavaScript is also used to display live map data, route information, ETA calculation, and interactive hospital recommendations.

Mapping Technologies

OpenStreetMap (OSM)

OpenStreetMap is used as the main map provider in the project. It provides map data for displaying roads, hospitals, ambulance locations, and navigation routes. OSM helps identify nearby hospitals and supports map visualization without requiring expensive commercial mapping services.

Overpass API

The Overpass API is used to fetch nearby hospital data from OpenStreetMap. It retrieves hospital details such as names, locations, emergency services, and healthcare facilities based on the ambulance's current GPS location.

OSRM (Open Source Routing Machine)

OSRM is used for route calculation and navigation. It provides shortest-route generation, travel distance estimation, and ETA calculation between the ambulance and selected hospital.

Leaflet.js

Leaflet.js is used to display interactive maps in the frontend. It integrates OpenStreetMap data with the web application and helps display ambulance tracking, hospital markers, live routes, and navigation paths visually on the map.

6.2 AI-Based Triage Implementation

- **Heart Rate:** (60–100 bpm)
- **Blood Pressure:** (90/60 – 120/80 mmHg)
- **Respiratory Rate:** (12–20 breaths/min)
- **SpO2:** (95%–100%)

- **Temperature:** (36.5°C – 37.5°C)
- **GCS:** (3–15 score)
- **Blood Glucose:** (70–140 mg/dL)
- **Symptoms:** (Chest pain, breathing difficulty, fever, unconsciousness, bleeding, etc.)

Example Conditions

Condition	**Result**
SpO2 < 90	Critical
GCS <= 8	Critical
Chest pain	High
Severe trauma	High

6.3 Distance Calculation

The project uses the **Haversine Formula** to calculate the distance between the ambulance and nearby hospitals based on their latitude and longitude. This helps in identifying the nearest hospital, estimating travel distance, and calculating the Estimated Time of Arrival (ETA), which improves emergency response and route optimization.

Haversine Formula:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2\left(\frac{\Delta\lambda}{2}\right)$$

$$c = 2 \cdot \tan^{-1}\left(\frac{\sqrt{a}}{\sqrt{1-a}}\right)$$

$$d = R \cdot c$$

6.4 Hospital Ranking Algorithm

The system uses a Hybrid Multi-Criteria Decision Making (MCDA) approach, which combines:

1. Rule-Based Filtering Algorithm

First, hospitals are filtered using mandatory conditions:

- ICU availability = Yes
- Emergency services = Available
- Trauma / Cardiac care support = Required check

2. Weighted Scoring Algorithm

After filtering, each hospital is assigned a score based on important parameters:

- Distance from ambulance (using Haversine formula)
- ICU availability
- Trauma care support

- Cardiac emergency support
- Emergency service availability
- Estimated Time of Arrival (ETA)

3. Haversine-Based Distance + ETA Calculation

- Distance is calculated using the Haversine Formula
- ETA is estimated using:

  Distance ÷ ambulance speed

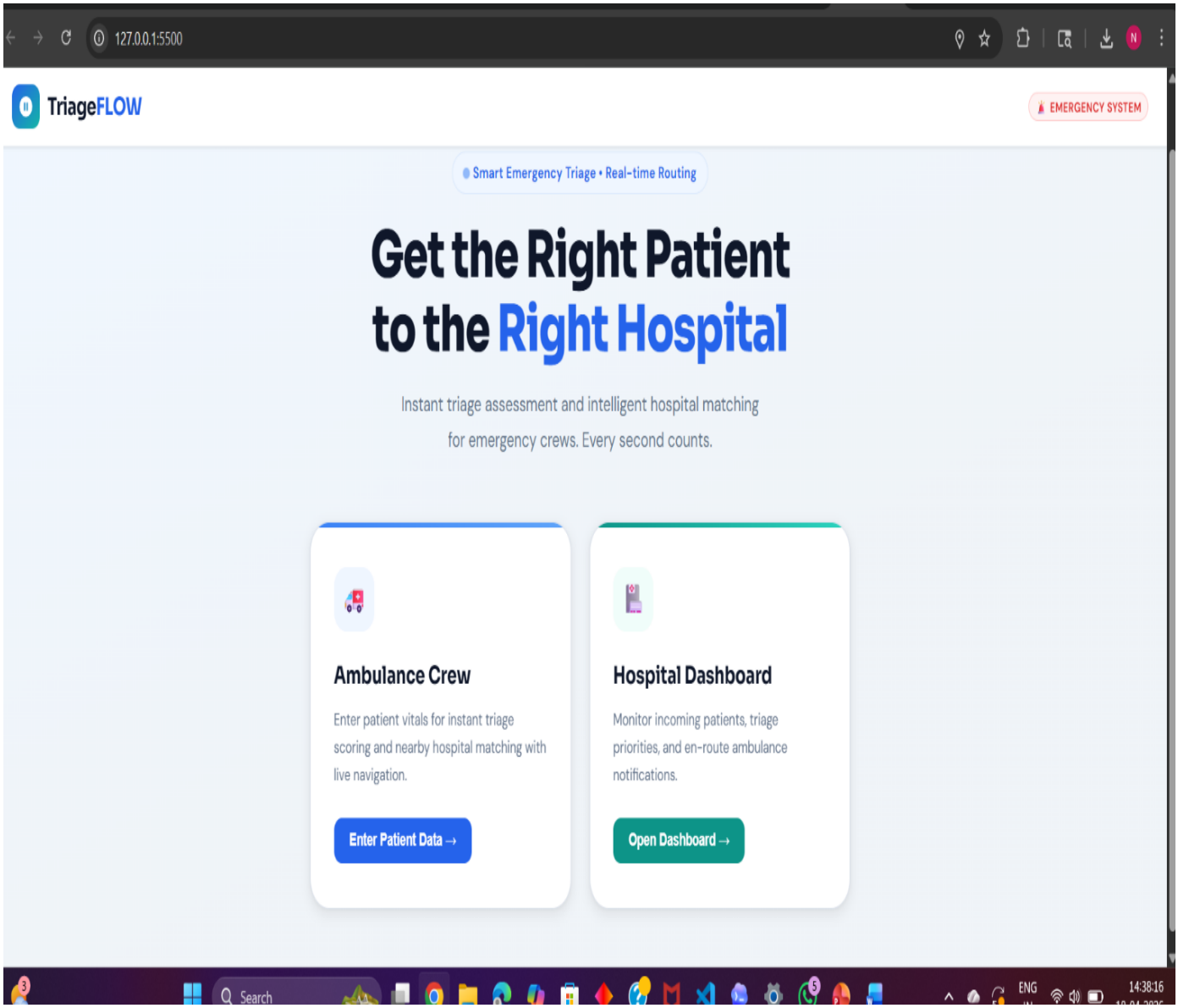
- ETA is used as a key ranking factor in scoring

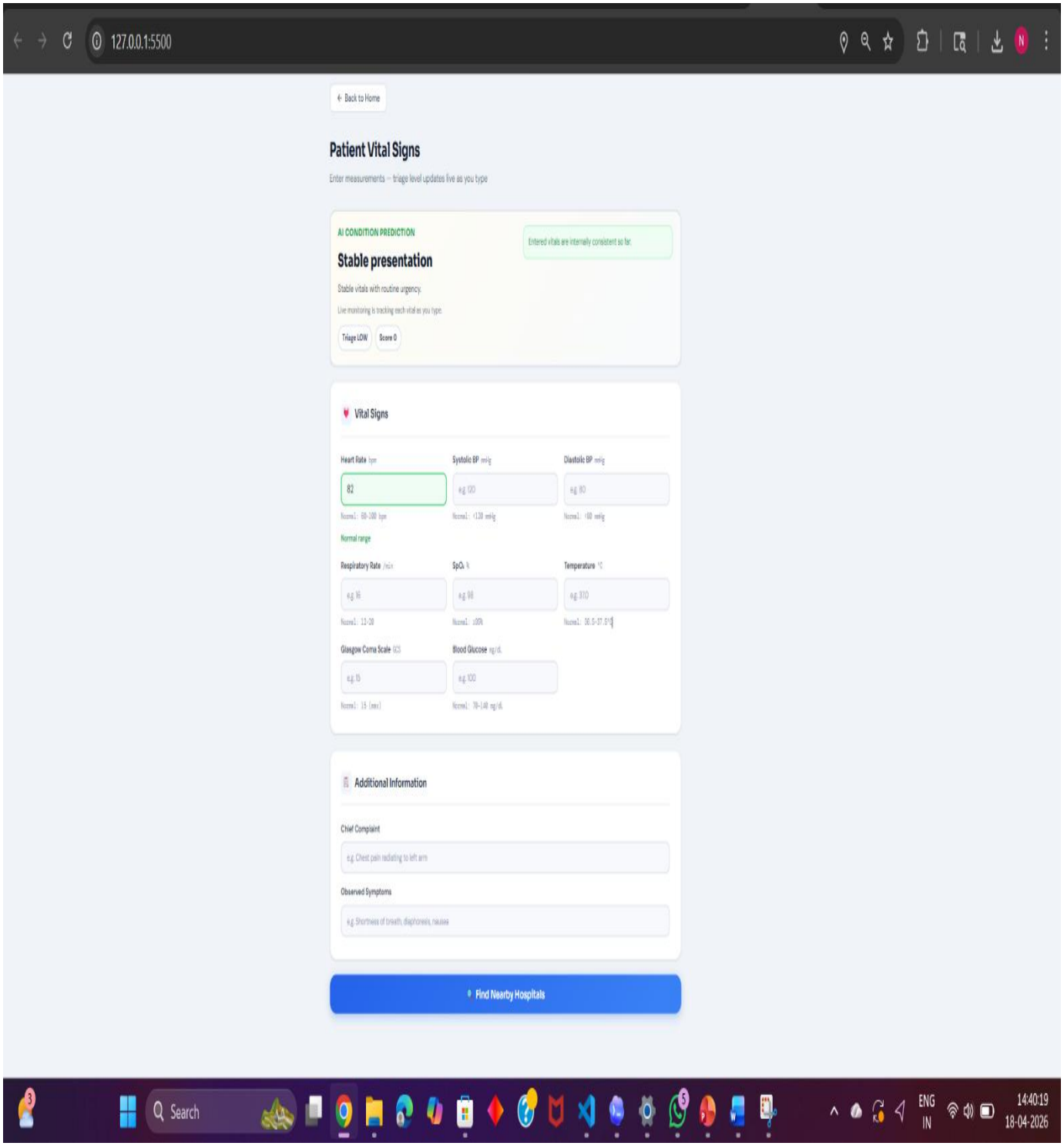
6.5 Route Navigation

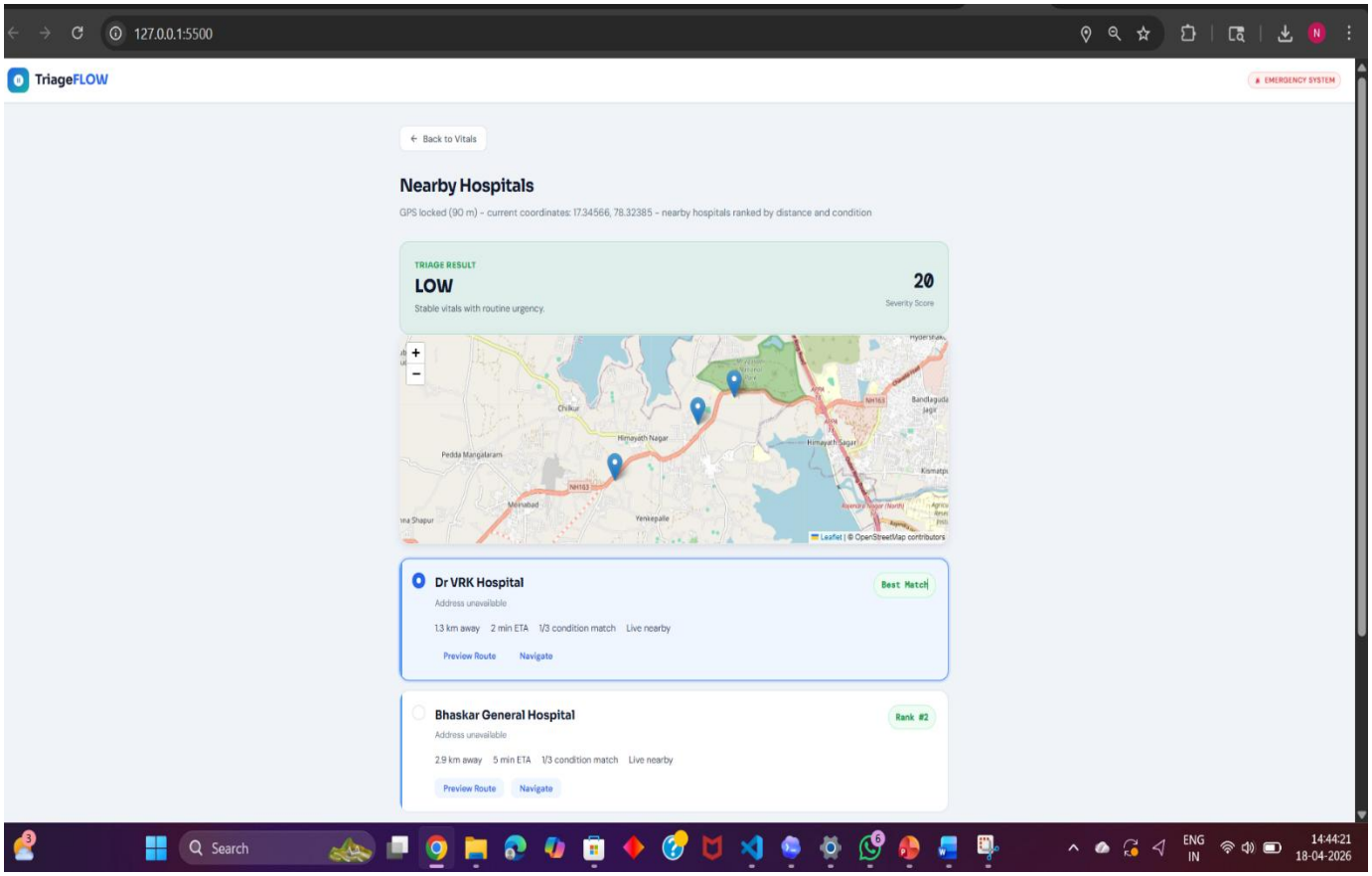
The routing module provides shortest route generation, live GPS navigation, distance estimation, and ETA (Estimated Time of Arrival) calculation. This helps ambulances reach hospitals quickly and reduces delays during emergency situations.

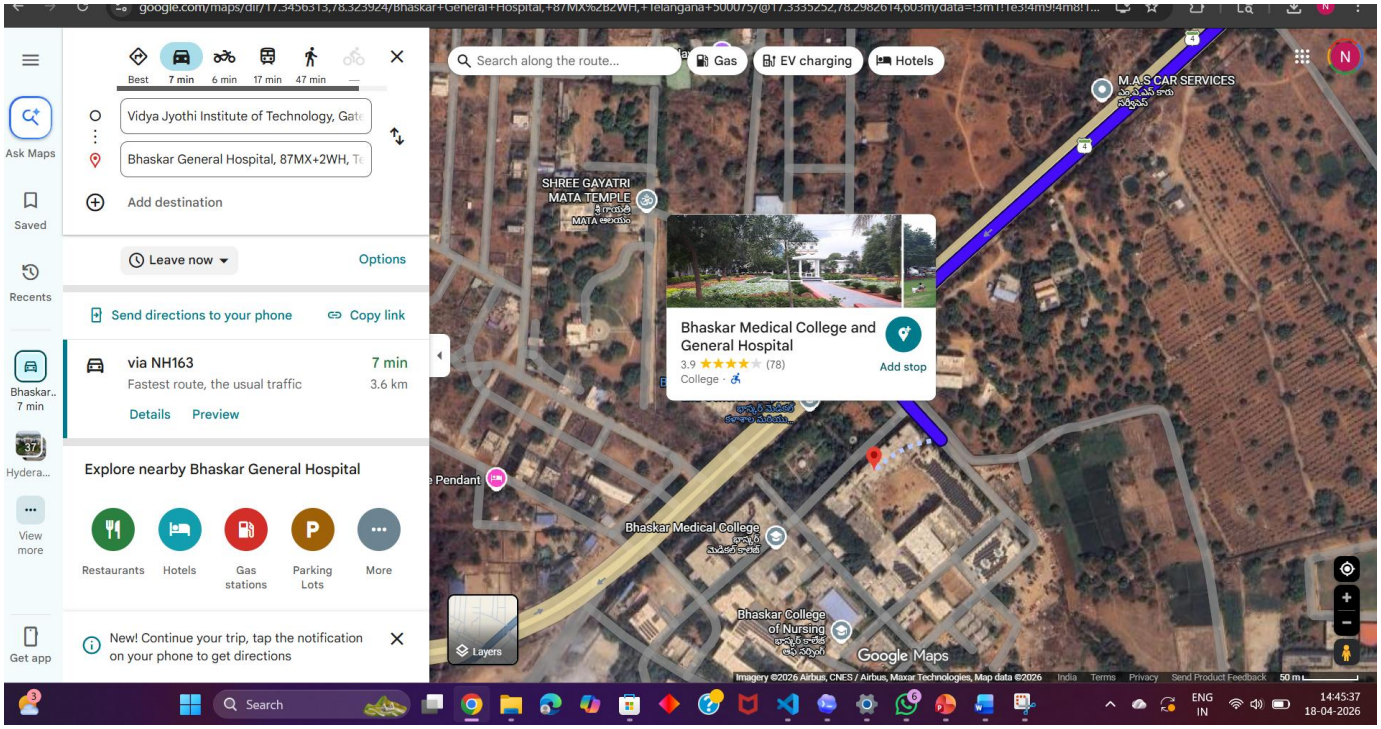
CHAPTER 7

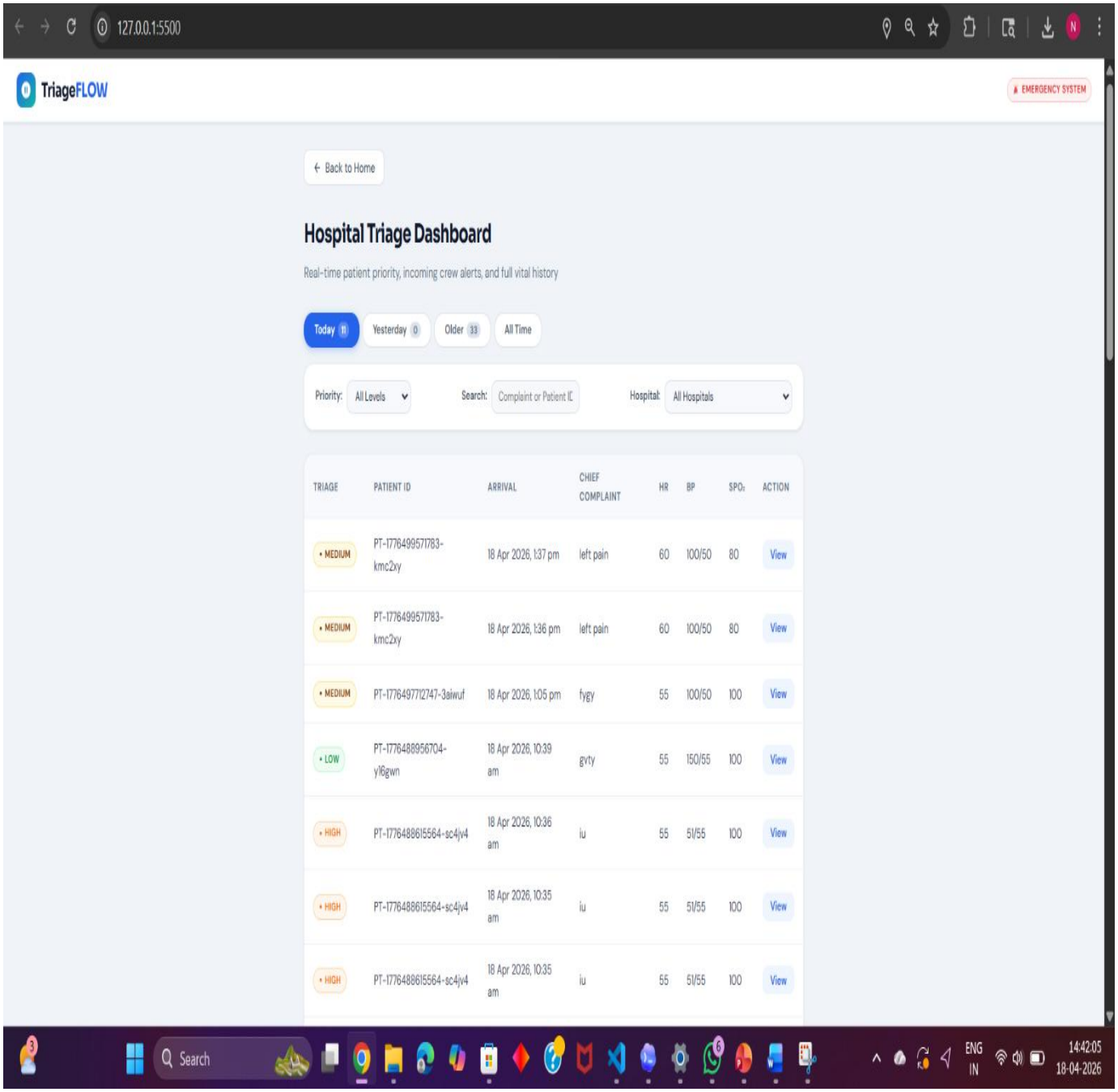
Result




127.0.0.1:5500
← Back to Home
Patient Vital Signs
Enter measurements — triage level updates live as you type
AI CONDITION PREDICTION
Entered vitals are internally consistent so far.
Stable presentation
Stable vitals with routine urgency.
Live monitoring is tracking each vital as you type.
Triage LOW
Score 0
Vital Signs
Heart Rate bpm
Syntolic BP mmHg
Diastolic BP mmHg
82
Normal range
Respiratory Rate /min
SpO₂ %
Temperature °C
Glasgow Coma Scale /15
Blood Glucose mg/dL
Additional Information
Chief Complaint
e.g. Chest pain radiating to left arm
Observed Symptoms
e.g. Shortness of breath, diaphoresis, nausea
Find Nearby Hospitals
Search
ENG
IN
14:40:19
18-04-2026


127.0.0.1:5500
TriageFLOW
EMERGENCY SYSTEM
Back to Vitals
Nearby Hospitals
GPS locked (90 m) - current coordinates: 17.34566, 78.32385 - nearby hospitals ranked by distance and condition
TRIAGE RESULT
LOW
Stable vitals with routine urgency.
20
Severity Score
Leaflet | © OpenStreetMap contributors
Dr VRK Hospital
Best Match
Address unavailable
1.3 km away 2 min ETA 1/3 condition match Live nearby
Preview Route
Navigate
Bhaskar General Hospital
Rank #2
Address unavailable
2.9 km away 5 min ETA 1/3 condition match Live nearby
Preview Route
Navigate
Search
ENG
IN
14:44:21
18-04-2026


Best
7 min
6 min
17 min
47 min
Vidya Jyothi Institute of Technology, Gate
Bhaskar General Hospital, 87MX+2WH,
Add destination
Leave now
Options
Send directions to your phone
Copy link
via NH163
7 min
Fastest route, the usual traffic
3.6 km
Details
Preview
Explore nearby Bhaskar General Hospital
Restaurants
Hotels
Gas stations
Parking Lots
More
New! Continue your trip, tap the notification on your phone to get directions
Ask Maps
Saved
Recents
Get app
Search along the route...
Gas
EV charging
Hotels
M.A.S CAR SERVICES
SHREE GAYATRI MATA TEMPLE
Bhaskar Medical College and General Hospital
3.9 (78)
College
Add stop
Bhaskar Medical College
Bhaskar College of Nursing
Layers
Google Maps
Search
ENG
IN
14:45:37
18-04-2026


127.0.0.1:5500
TriageFLOW
EMERGENCY SYSTEM
Back to Home
Hospital Triage Dashboard
Real-time patient priority, incoming crew alerts, and full vital history
Today 11
Yesterday 0
Older 33
All Time
Priority: All Levels
Search: Complaint or Patient ID
Hospital: All Hospitals
TRIAGE
PATIENT ID
ARRIVAL
CHIEF COMPLAINT
HR
BP
SPO2
ACTION
MEDIUM
PT-1776499571783-kmc2xy
18 Apr 2026, 1:37 pm
left pain
60
100/50
80
View
MEDIUM
PT-1776499571783-kmc2xy
18 Apr 2026, 1:36 pm
left pain
60
100/50
80
View
MEDIUM
PT-1776497712747-3aiwuf
18 Apr 2026, 1:05 pm
fygy
55
100/50
100
View
LOW
PT-1776488956704-yl6gwn
18 Apr 2026, 10:39 am
gvty
55
150/55
100
View
HIGH
PT-1776488615564-sc4jv4
18 Apr 2026, 10:36 am
iu
55
51/55
100
View
HIGH
PT-1776488615564-sc4jv4
18 Apr 2026, 10:35 am
iu
55
51/55
100
View
HIGH
PT-1776488615564-sc4jv4
18 Apr 2026, 10:35 am
iu
55
51/55
100
View
Search
ENG
IN
14:42:05
18-04-2026

CHAPTER 8
TESTING

8.1 Introduction

Testing is a crucial phase in the development of the system to ensure that all components work correctly and efficiently under different conditions. It helps in identifying bugs, verifying system requirements, and ensuring reliability and accuracy. In the TriageFLOW system, testing ensures that emergency classification, GPS tracking, hospital selection, and route optimization work as expected in real-time scenarios.

The main goal of testing is to ensure:

- Accuracy of patient triage classification
- Proper functioning of GPS-based hospital detection
- Correct calculation of distance and ETA
- Reliable communication between modules

8.2 Types of Testing

1. Unit Testing

- Each individual module or function is tested separately.
- Ensures that every component works correctly in isolation.

Example:

  - Testing Haversine formula function for correct distance output
  - Testing severity calculation based on vitals (SpO_2, BP, HR)

2. Integration Testing

- Multiple modules are combined and tested together.
- Ensures data flows correctly between components.

Example:

  - Patient input module + AI triage system + hospital selection system
  - GPS module integrated with route optimization system

3. Functional Testing

- Validates the complete system against requirements.
- Checks whether the system produces correct outputs for given inputs.

Example:

  - Entering symptoms should correctly classify emergency level
  - Selecting a hospital should generate navigation route

4. System Testing (Optional but important)

- Entire system is tested as a whole in a real-world environment.
- Ensures performance, reliability, and usability.

Example:

  - Simulating emergency cases with live GPS tracking and hospital response

5. Performance Testing

- Checks system response time and efficiency.
- Ensures fast processing for emergency situations.

Example:

  - Time taken to compute nearest hospital and ETA

8.3 Test Cases

Test Case	Input	Expected Output	Result
Low SpO_2 Level	85	Critical alert triggered with emergency notification	Pass
GPS Enabled	Location access granted	Nearby hospital list displayed based on distance	Pass
Chest Pain Symptoms	Chest pain, dizziness	High severity classification (Red Triage Level)	Pass
Route Request	Hospital selected	Navigation route generated with shortest path	Pass
Normal Vitals	HR: 75, SpO_2: 98	Low priority / Green category	Pass
Distance Calculation	Ambulance & hospital coordinates	Accurate distance using Haversine formula	Pass

CHAPTER 9

CONCLUSIONAND FUTURE ENHANCEMENTS

9.1 CONCLUSION

TriageFLOW is an intelligent emergency healthcare and ambulance navigation system developed to improve emergency response efficiency and patient care during critical situations. The project combines AI-based triage analysis, GPS tracking, hospital recommendation, and real-time navigation into a single smart healthcare platform.

The system analyzes patient vital signs such as heart rate, blood pressure, SpO2, temperature, GCS score, and symptoms to determine emergency severity levels and predict possible medical conditions. Based on the patient's condition and current location, TriageFLOW identifies nearby hospitals using OpenStreetMap and Overpass APIs and ranks them according to factors such as ICU availability, trauma care support, emergency facilities, distance, and estimated arrival time (ETA).

The routing and navigation module helps ambulances reach hospitals quickly by providing shortest-route generation, live GPS tracking, and real-time ETA updates. This reduces transportation delays and improves coordination between ambulance services and hospitals. The system also supports better patient monitoring, emergency handling, and healthcare communication during critical situations.

9.2 Future Enhancements

Machine Learning Integration

Future versions of TriageFLOW can use machine learning algorithms for more accurate patient condition prediction and intelligent emergency analysis based on historical medical data.

Cloud Database Support

Cloud-based database integration can improve data storage, backup, scalability, and real-time access to patient records and hospital information from multiple locations.

Mobile Application

A dedicated mobile application can make the system more accessible for ambulance staff, doctors, and emergency responders using smartphones and tablets.

IoT Healthcare Sensors

The system can be integrated with IoT-based healthcare sensors to automatically collect real-time patient vital signs such as heart rate, blood pressure, temperature, and SpO2.

Voice Assistant Support

Voice assistant features can help ambulance staff operate the system hands-free during emergencies using voice commands for navigation and patient updates.

Traffic Prediction AI

AI-based traffic prediction can help identify traffic congestion in advance and suggest faster alternate routes for ambulances.

Real-Time Hospital Bed Availability

Future integration with hospital databases can provide real-time information about ICU beds, emergency wards, ventilators, and doctor availability for better hospital selection

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